No of Pages : 3 **Course Code : 23MX13**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, JUNE 2024

MCA Semester: 1

23MX13 DATA STRUCTURES

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.
2. **If any information is missing, make appropriate assumptions and mention the same.**

1. a) Define asymptotic notations for lower and upper bound complexity of algorithms. What is the complexity of the following code? (5)

```
sum = 0
for ( i = 0; i < n/2; i ++)
    for ( j = 1; j < 8 + i ; j ++)
        for ( k = 0; k < 8 ; k ++)
            print("$")
```

b) i) What are the properties of an algorithm? Given an array of size N-1 with distinct integers in the range of 1 to N, write an algorithm to find the missing element. What is the time and space complexity of your algorithm? (4)

ii) For a task, different algorithms are written and their complexity are O(n), O(log n), O(nlog n), O(n^2), O(2^n). Which is good and which is worst? Justify your answer. (4)

c) i) Solve the recurrence relation T(n) = T(n/2) + n (6)

ii) Write the recurrence relation corresponding to the following algorithm. Solve the same to find the complexity. (6)

```
Algorithm TOH (n, x, y, z)
if (n >= 1)    {
    TOH ( (n-1), x, z, y)

    x = y
    TOH ((n-1), z, y, x)
}
```

2. a) Write an algorithm to find the sum of the elements stored in a linked list. Insert a node at the end of the list and store the sum. (5)

   b) i) A 2- dimensional array A(5 x 4) of integers is stored starting from location 1000. An integer occupies 2 bytes. Calculate the address of the element A[3, 2] if the array is stored using i. row major order and ii. Column major order. Write the addressing functions. (4)

   ii) What are sparse matrices? Suggest an efficient storage representation of sparse matrix. Illustrate with an example. (4)

   c) i) Consider a mathematical expression with ( ) , [ ] and { }. Parentheses may be nested. Write an algorithm to check for the well formedness of parentheses in the given expression. (6)

   ii) Write algorithms for the primitive operations on queue. (6)

3. a) How are binary trees represented in storage? Illustrate with suitable example. Which type of representation is efficient for i. complete binary tree and ii. Skew tree? Justify your answer. (5)

   b) i) What are the different ways of traversing a binary tree? Indicate the order in which the nodes are visited. Write the traversal sequences for the tree shown in Fig 1. (6)

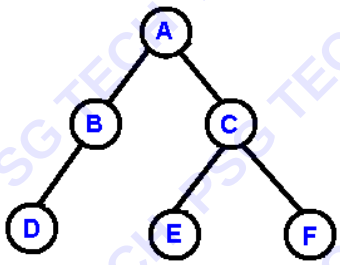

Fig 1

   ii) How are graphs represented in storage? Show the representations for the graph shown in Fig 2.Indicate the advantage and limitation of each representation. (6)

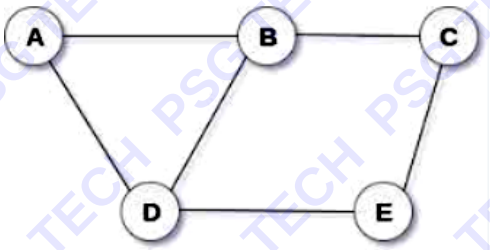

Fig 2

c) i) What is a Binary Serach Tree? Construct a Binary Search Tree by inserting the elements 56, 34, 99, 12, 39, 78, 45. Delete 34 and shor the resultatnt tree. Write algorithms for isnertion and deletion operation.

(OR)

ii) Write algorithms for Breadth first and depth first traversal on graphs. Trace the algorithms on the graph shown in Fig 2.

4. a) Sort the following list of elements using Radix Sort : 56, 34, 99, 12, 39, 78, 45, 5, 122. What is the complexity of Radix Sort. (5)

b) Write algorithm for linear and binary Search. Indicate the best and worst case complexity. Illustrate with suitable examples. (8)

c) i) 1) Write algorithm for Insertion Sort. What is its complexity? Sort 56, 34, 99, 12, 39, 78, 45 and show the results of intermediates passes also. (6)

2) Sort the lsit 56, 34, 99, 12, 39, 78, 45, 5, 122 using heap sor. What is the complexity of heap sort? (6)

(OR)

ii) Consider hash function for n is n % 20 and second hash function, where applicable, is

(n % 7) + 1. Suppose separate chaining is used for collision resolution. Beginning with an empty hash table, insert the following elements 8, 38, 3, 5, 28, 18, 65, 83 into the hash table. How many of the inserted keys will be looked at when 48 is searched?

1) Suppose linear probing is used for collision resolution. What is the answer to the above questions?

2) Suppose double hashing is used for collision resolution. What is the answer to the above questions?

/END/

FD/RSP